DDA2001: Introduction to Data Science

Final Exam
SDS, CUHK(SZ)

May 13, 2025

Instructions

1. The time duration is **120 minutes**.
2. The total mark is **100 points**.
3. There are **7 questions** in total, please check.
4. Please write down your calculations and statements **clearly and in detail**.
5. Give answers up to **2 decimal places** where appropriate.
6. You should answer the questions in the answer sheet in **sequential order**.
7. Please submit the test paper and answer sheet **together** at the end.

Declare: I have read the instructions carefully. I will fully comply with the exam rules and instructions.

Name: _____ Student ID: _____

Useful Formulas.

- Let X be a random variable. Define $f(y) = \mathbb{E}[\min(y, X)]$. Then

$$f'(y) = \mathbb{P}(X \geq y).$$

- Mean and variance for common distributions

Name of the probability distribution	Probability distribution function	Mean	Variance
Binomial distribution	$\Pr(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$
Geometric distribution	$\Pr(X = k) = (1-p)^{k-1} p$	$\frac{1}{p}$	$\frac{(1-p)}{p^2}$
Normal distribution	$f\left(x \mid \mu, \sigma^2\right) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2
Uniform distribution (continuous)	$f(x \mid a, b) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b, \\ 0 & \text{for } x < a \text{ or } x > b \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$

- When $\theta = (\theta_1, \cdots, \theta_n)$ and $z = (z_1, \cdots, z_n)$, $\theta^T z = \sum_{i=1}^n \theta_i z_i$.

1. **(10 points) K-Means**

   (a) **(3 points)** Explain the process of clustering a set of data into K segments using the K-Means algorithm.

   (b) **(3 points)** Describe the three key properties that a distance function used in the K-Means algorithm should satisfy.

   (c) **(4 points)** An online store tracks the purchase frequency and spending of its customers. Consider a dataset comprising 6 customers, represented as follows: $(5,5), (6,8), (4,2), (20,15), (16,20), (25,10)$. We initialize the first two cluster centers as $c_1^{(0)} = (5,5)$ and $c_2^{(0)} = (20,15)$. Calculate the next pair of cluster centers $c_1^{(1)}$ and $c_2^{(1)}$ (use Euclidean distance).

Solution

(a) As long as the provided answer covers the following steps, we can give the full credit: (1) cluster assignment; (2) cluster center update; (3) stop until no assignment and center update.

(b) Symmetry, positive separability, and triangular inequality.

(c) For each point, compute the Euclidean distance to both cluster centers and assign it to the nearest one.

- **Point (5, 5)**: Closest to $c_1^{(0)}$.
- **Point (6, 8)**: Distance to $c_1^{(0)}$ is $\sqrt{(6-5)^2+(8-5)^2} = \sqrt{10}$; closer to $c_1^{(0)}$.
- **Point (4, 2)**: Distance to $c_1^{(0)}$ is $\sqrt{(4-5)^2+(2-5)^2} = \sqrt{10}$; closer to $c_1^{(0)}$.
- **Point (20, 15)**: Closest to $c_2^{(0)}$.
- **Point (16, 20)**: Distance to $c_2^{(0)}$ is $\sqrt{(16-20)^2+(20-15)^2} = \sqrt{41}$; closer to $c_2^{(0)}$.
- **Point (25, 10)**: Distance to $c_2^{(0)}$ is $\sqrt{(25-20)^2+(10-15)^2} = \sqrt{50}$; closer to $c_2^{(0)}$.

Cluster 1 (assigned to $c_1^{(0)}$): (5, 5), (6, 8), (4, 2). **Cluster 2 (assigned to $c_2^{(0)}$)**: (20, 15), (16, 20), (25, 10). The new cluster centers are the means of the points in each cluster.

- **New center for Cluster 1 ($c_1^{(1)}$)**:

$$\begin{aligned}\text{Mean of } x\text{-coordinates} &= \frac{5+6+4}{3} = 5\\ \text{Mean of } y\text{-coordinates} &= \frac{5+8+2}{3} = 5\\ \Rightarrow c_1^{(1)} &= (5,5)\end{aligned}$$

- **New center for Cluster 2 ($c_2^{(1)}$)**:

$$\begin{aligned}\text{Mean of } x\text{-coordinates} &= \frac{20+16+25}{3} = \frac{61}{3}\\ \text{Mean of } y\text{-coordinates} &= \frac{15+20+10}{3} = 15\\ \Rightarrow c_2^{(1)} &= \left(\frac{61}{3}, 15\right)\end{aligned}$$

The next pair of cluster centers are:

$$\boxed{(5,5)} \quad \text{and} \quad \boxed{\left(\frac{61}{3}, 15\right)}.$$

2. **(10 points) Convex Sets**

(a) **(5 points)** Let $\Delta_n = \{\mathbf{p} \in \mathbb{R}^n | p_i \geq 0, \sum_{i=1}^{n} p_i = 1\}$, where $\mathbf{p} = (p_1, ..., p_n)$. Show that Δ_n is a convex set.

(b) **(5 points)** Show that the subset $S = \{\mathbf{p} \in \Delta_n | -\sum_{i=1}^{n} p_i \ln p_i \geq C\}$ is convex for any $C \in \mathbb{R}$, where Δ_n is defined in part (a).

Solution

(a) Take any two points $\boldsymbol{p}, \boldsymbol{q} \in \Delta_n$ and $\lambda \in [0, 1]$. We must show that $\lambda\boldsymbol{p} + (1-\lambda)\boldsymbol{q} \in \Delta_n$. Since $p_i, q_i \geq 0$ and $\lambda \in [0, 1]$, each component satisfies:

$$(\lambda\boldsymbol{p} + (1-\lambda)\boldsymbol{q})_i = \lambda p_i + (1-\lambda)q_i \geq 0.$$

Moreover,

$$\sum_{i=1}^{n}(\lambda\boldsymbol{p} + (1-\lambda)\boldsymbol{q})_i = \lambda\sum_{i=1}^{n} p_i + (1-\lambda)\sum_{i=1}^{n} q_i = \lambda \cdot 1 + (1-\lambda) \cdot 1 = 1.$$

Thus, $\lambda\boldsymbol{p} + (1-\lambda)\boldsymbol{q} \in \Delta_n$, so Δ_n is convex.

(b) Δ_n is convex (from part (i)). S is the intersection of Δ_n and the sub-level set $\{\boldsymbol{p} \mid \sum_{i=1}^{n} p_i \ln p_i \leq -C\}$, which is convex as the function $f(\boldsymbol{p}) = \sum_{i=1}^{n} p_i \ln p_i$ is a convex function.

3. **(15 points) Logistic Regression with Polynomial Features**

Suppose you are given a binary classification task where each input $x_i \in \mathbb{R}^2$, and the feature vector $\phi(x_i)$ includes all monomials up to degree 2. That is, for $x_i = (x_{i1}, x_{i2})$, we define

$$\phi(x_i) = (1, x_{i1}, x_{i2}, x_{i1}^2, x_{i2}^2, x_{i1}x_{i2})^\top \in \mathbb{R}^6.$$

The model is

$$\mathbb{P}(y_i = 1 \mid x_i, \theta) = \frac{1}{1 + e^{-\theta^\top \phi(x_i)}},$$

where $y_i \in \{0, 1\}$ and $\theta \in \mathbb{R}^6$ are the model parameters.

(a) **(4 points)** Derive the log-likelihood function $\ell(\theta)$ for n independent training examples $\{(x_i, y_i)\}_{i=1}^n$. Write $\ell(\theta)$ explicitly in terms of $\phi(x_i)$ and θ.

(b) **(4 points)** Is the negative log-likelihood function $-\ell(\theta)$ convex in θ? Explain carefully.

(c) **(5 points)** Suppose you want to find θ by gradient descent. Write the update rule using a learning rate $\eta > 0$. How does stochastic gradient descent modify the update rule?

(d) **(2 points)** Describe qualitatively the shape of the decision boundary that logistic regression learns in this case. How does it compare to standard logistic regression without polynomial features?

Solution

(a) The likelihood for n independent training examples (x_i, y_i), $i = 1, \ldots, n$, is

$$L(\theta) = \prod_{i=1}^{n} \sigma(\theta^\top \phi(x_i))^{y_i} (1 - \sigma(\theta^\top \phi(x_i)))^{1-y_i},$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function. Taking the logarithm of the likelihood, we get the log-likelihood:

$$\ell(\theta) = \sum_{i=1}^{n} \left[y_i \log \sigma(\theta^\top \phi(x_i)) + (1 - y_i) \log(1 - \sigma(\theta^\top \phi(x_i))) \right].$$

Now simplify:

$$\ell(\theta) = \sum_{i=1}^{n} \left[y_i \theta^\top \phi(x_i) - \log(1 + e^{\theta^\top \phi(x_i)}) \right].$$

Thus, the log-likelihood function is:

$$\boxed{\ell(\theta) = \sum_{i=1}^{n} \left(y_i \theta^\top \phi(x_i) - \log\left(1 + e^{\theta^\top \phi(x_i)}\right) \right)}.$$

(b) TWe want to check if $-\ell(\theta)$ is convex in θ. Since the function $\log(1+e^z)$ is convex (second order condition) and the term $\theta^\top \phi(x_i)$ is affine (linear), the negative log-likelihood is a sum of convex functions. Therefore, $-\ell(\theta)$ is convex in θ.

$$\boxed{\text{Yes, } -\ell(\theta) \text{ is convex in } \theta.}$$

(c) We first compute the gradient of the log-likelihood:

$$\nabla_\theta \ell(\theta) = \sum_{i=1}^{n} (y_i - \sigma(\theta^\top \phi(x_i))) \phi(x_i),$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function. The update rule for gradient descent is:

$$\theta^{(t+1)} = \theta^{(t)} + \eta \nabla_\theta \ell(\theta^{(t)}).$$

Explicitly, this becomes:

$$\boxed{\theta^{(t+1)} = \theta^{(t)} + \eta \frac{1}{n} \sum_{i=1}^{n} (y_i - \sigma(\theta^{(t)\top} \phi(x_i))) \phi(x_i)}.$$

For stochastic gradient descent (SGD), we update θ using a mini-batch of training examples. Let B be the mini-batch of size m, where $B = \{(x_{i1}, y_{i1}), \ldots, (x_{im}, y_{im})\}$,

a random subset of the n training points. The update rule for mini-batch SGD is:

$$\boxed{\theta^{(t+1)} = \theta^{(t)} + \eta \frac{1}{m} \sum_{i \in B} (y_i - \sigma(\theta^{(t)\top} \phi(x_i))) \phi(x_i)}.$$

Here, B represents the mini-batch, and at each step, we calculate the gradient over the subset of data points in B instead of the full dataset.

(d) The decision boundary is where $\mathbb{P}(y = 1 \mid x) = 0.5$, which occurs when:

$$\theta^\top \phi(x) = 0.$$

Explicitly, this gives the decision boundary equation:

$$\theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_1^2 + \theta_4 x_2^2 + \theta_5 x_1 x_2 = 0.$$

Thus, the decision boundary is a quadratic curve in (x_1, x_2).

In contrast, standard logistic regression without polynomial features fits a linear decision boundary.

Therefore, with polynomial features, the decision boundary can be more complex and curved (the key is to mention **nonlinear**), unlike the **linear** decision boundary in standard logistic regression.

4. **(10 points) Gradient Descent Method**

Let's maximize the function $f(x) = -\frac{1}{4}(x-1)^2 + 2$ using the gradient descent method. We fix the learning rate in the algorithm as a constant, α. Assume that the initial value of x is $x^{(0)} = 3$, and $x^{(t)}$ denotes the value of x after the t-th iteration of gradient descent.

(a) In each of the following cases, will $\lim_{t\to\infty} x^{(t)}$ converge to the optimal solution? Justify your answer.

i. **(2 point)** $\alpha = 0$.

ii. **(2 point)** $\alpha = 4$.

iii. **(2 point)** $\alpha = 1$.

(b) **(4 points)** For what range of α will $\lim_{t\to\infty} x^{(t)}$ converge to the optimal solution? Explain your reasoning. (Hint: derive the update rule and analyze convergence using the recursion.)

Solution: Let's maximize the function $f(x) = -\frac{1}{4}(x-1)^2 + 2$ using the gradient descent method (equivalently, gradient ascent in this case, since we are maximizing). We fix the learning rate in the algorithm as a constant, α. Assume that the initial value of x is $x^{(0)} = 3$, and $x^{(t)}$ denotes the value of x after the t-th iteration of gradient descent. The stopping rule is $|x^{(t+1)} - x^{(t)}| \leq \epsilon$, where ϵ is a non-negative constant.

(a) Let $\epsilon = \frac{1}{1000}$. In each of the following cases, will the final output be the optimal solution? Justify your answer.

i. $\alpha = 0$

The update rule is:

$$x^{(t+1)} = x^{(t)} + \alpha f'(x^{(t)})$$

Since $\alpha = 0$, there is no update and $x^{(t+1)} = x^{(t)}$ for all t. Therefore, the algorithm stays at $x^{(0)} = 3$ and does not move toward the optimum.

Answer: No, the final output will not be the optimal solution.

ii. $\alpha = 4$

First, compute the gradient:

$$f'(x) = -\frac{1}{2}(x-1)$$

The update becomes:

$$x^{(t+1)} = x^{(t)} - \frac{\alpha}{2}(x^{(t)} - 1)$$

Define the error $e^{(t)} = x^{(t)} - 1$. Then:

$$e^{(t+1)} = \left(1 - \frac{\alpha}{2}\right) e^{(t)}$$

Substituting $\alpha = 4$:

$$e^{(t+1)} = (1-2)e^{(t)} = -e^{(t)}$$

Hence, $e^{(t)}$ oscillates between ± 2 and never converges.

Answer: No, the algorithm oscillates and does not converge to the optimum.

iii. $\alpha = 1$

$$e^{(t+1)} = \left(1 - \frac{1}{2}\right) e^{(t)} = \frac{1}{2} e^{(t)}$$

The error decays geometrically, so $x^{(t)} \to 1$.

Answer: Yes, the algorithm converges to the optimal solution.

(b) Let $\epsilon = 0$. For what range of α will $\lim_{t\to\infty} x^{(t)}$ converge to the optimal solution? Explain your reasoning.

We analyze the convergence of the recursion:

$$e^{(t+1)} = \left(1 - \frac{\alpha}{2}\right) e^{(t)}$$

For convergence, we need:

$$\left|1 - \frac{\alpha}{2}\right| < 1 \Rightarrow -1 < 1 - \frac{\alpha}{2} < 1 \Rightarrow 0 < \alpha < 4$$

Answer: The algorithm converges to the optimal solution if and only if $\boxed{0 < \alpha < 4}$.

5. **(10 points) Convex or Concave Functions**

   (a) **(5 points)** Suppose $f : \mathbb{R}^n \to \mathbb{R}$ is concave, with $\operatorname{dom} f = \mathbb{R}^n$, and bounded below (i.e., there exists a constant $M < 0$ such that $f(x) \geq M$ for all $x \in \mathbb{R}^n$). Show that f must be a constant function.

   (b) **(5 points)** For what range of a will function $f(x) = x^{a^2-a} + a \cdot x$ be a real-valued concave function within interval (a^2, ∞)? Explain the reasons.

   Solution

   (a) See exercise 5, lecture 16.

   (b) As $f''(x) = (a^2-a)(a^2-a-1)x^{a^2-a-2}$ and $x^{a^2-a-2} \geq 0$ for any $x \geq a^2$, the desired condition reduces to

   $$(a^2 - a)(a^2 - a - 1) \leq 0$$

   Thus $a \in [\frac{1-\sqrt{5}}{2}, 0] \cup [1, \frac{1+\sqrt{5}}{2}]$.

6. **(26 points) Probability + Statistics + Optimization + Machine Learning**
   A ride-hailing company is analyzing its data to improve platform performance.

   (a) **Ride Acceptance modelling (11 points)** The company tests user response to prices. A user is shown a price p, and accepts the ride ($Y = 1$) with probability

   $$\theta(p) = e^{-\lambda p}, \quad \text{with } \lambda > 0.$$

      i. **(3 points)** Suppose n users are shown the same price p. Let S be the number of accepted rides. What is the distribution of S? What are $\mathbb{E}[S]$ and $\mathrm{Var}[S]$?

      ii. **(2 points)** In the experiments, prices are drawn from $p \sim$Uniform$[10, 20]$. Compute $\mathbb{E}[p]$ and $\mathrm{Var}[p]$.

      iii. **(3 points)** Given observations $(p_1, y_2), ..., (p_n, y_n)$ with $y_i \in \{0, 1\}$, write the log-likelihood function for estimating λ.

      iv. **(3 points)** With the above log-likelihood function, how to find the value of the maximum likelihood estimator $\hat{\lambda}$ and why?

   (b) **Optimizing Price (7 points)** The company wants to find the optimal price p that maximizes the expected revenue $R(p) = p\theta(p)$.

      i. **(3 points)** Show that $R(p)$ is concave on $p \in [0, \frac{2}{\lambda}]$.

      ii. **(4 points)** Suppose the estimated value of λ from the data is $\lambda = 0.2$. Determine the optimal price $p^* \in [0, 10]$. What is the expected revenue at this price?

   (c) **Behavioral Segmentation (8 points)**
   The company collects the following features for each user i. V_i: average rides/week; D_i: average travel distance (km). The data for six users is shown below.

Table 1: User ride behavior data

User	Rides/week (V)	Distance (km) (D)
A	6	3
B	5	2
C	1	10
D	2	11
E	6	2
F	2	9

i. **(5 points)** Suppose the company uses K-means clustering with $K = 2$, initialized with centers at user A and user C. Perform one iteration of clustering: assign each user to the nearest center using Euclidean distance. Then compute the new centers of each cluster.

ii. **(3 points)** A new user arrives with profile $V, D = (5, 3)$. To which cluster should she be assigned?

Solution

(a) i. S is Binomial$(n, \theta(p))$. The mean is $\mathbb{E}[S] = ne^{-\lambda p}$. The variance is $\text{Var}[S] = ne^{-\lambda p}(1 - e^{-\lambda p})$.

ii. The mean is $\mathbb{E}[p] = 15$. The variance is $\text{Var}[p] = \frac{100}{12} \approx 8.33$.

iii. Since $\theta(p_i) = e^{-\lambda p_i}$, the log-likelihood is

$$\begin{aligned} l(\lambda) &= \sum_{i=1}^{n} \left[y_i \log(e^{-\lambda p_i}) + (1 - y_i) \log(1 - e^{-\lambda p_i}) \right] \\ &= \sum_{i=1}^{n} \left[-\lambda p_i y_i + (1 - y_i) \log(1 - e^{-\lambda p_i}) \right]. \end{aligned}$$

iv. One can find the value of the MLE using gradient ascent, since the log-likelihood function is concave by the second-order condition. By gradient ascent (or descent, both are fine), we update λ_t by

$$\lambda_{t+1} = \lambda_t + \eta * l'(\lambda_t)$$

where η is the step size.

(b) i. Use the second-order condition.

$$R'(p) = e^{-\lambda p}(1 - \lambda p), \quad R''(p) = -\lambda e^{-\lambda p}(2 - \lambda p).$$

So $R''(p) \leq 0$ when $\lambda \leq \frac{2}{\lambda}$, hence R is concave on $[0, \frac{2}{\lambda}]$.

ii. Use the first-order condition:

$$R'(p^*) = 0$$

gives $p^* = \frac{1}{\lambda} = 5$. The expected revenue at this price is $R(5) = 5/e \approx 1.839$.

(c) i. Initial centers: Cluster 1: A $(6, 3)$; Cluster 2: B $(1, 10)$.
Assignments: ABE to 1, and CDF to 2.
New centers: Cluster 1: $(5.67, 2.33)$; Cluster 2: $(1.67, 10)$.

ii. It is closest to Cluster 1 and should be assigned to Cluster 1.

iii. Offer ride bundles, subscription plans, or loyalty discounts to encourage continued high-frequency use.

7. **(19 points) MLE + Convex Optimization**
A seller plans to sell a single type of product and faces uncertain demand D. To meet the demand, the seller can produce units in two ways:

- Pre-production: Before sales begin, the seller can produce any number of units at a cost of \$5 per unit. However, any unsold pre-produced units will have no salvage value (i.e., zero residual value).
- Rush production: After observing the realized demand D, the seller can rush produce additional units if needed, but at a higher cost of \$9 per unit.

The demand D can be satisfied either by pre-produced units or rush-produced units, where each unit sold generates a revenue of $\$p$, where $p > 9$. Pre-production offers a lower cost but carries the risk of overproducing, while rush production is more flexible but incurs a higher cost. The seller must determine the optimal pre-production quantity before observing the realized demand, with the objective of maximizing expected profit.

(a) (3 points) After observing the realized demand D, how many units should the seller rush produce? Prove that the optimal number of rush-produced units is $(D - Q)^+$, where $x^+ = \max(x, 0)$. Explain your reasoning.

(b) (3 points) Show that at the optimal decision, the seller's total sales revenue (i.e., the total amount collected from selling products, excluding any production costs) is $p\mathbb{E}[D]$. That is, all demand will be satisfied.

(c) (4 points) Suppose the random demand D follows a **uniform distribution** over $[\theta, \theta^2]$, where θ is an unknown number. You are given 5 observed samples:

$$\{12,\ 15,\ 13,\ 16,\ 14\}$$

Find the MLE of θ. Explain your steps clearly.

(d) (6 points) Now assume that D follows a uniform distribution over $[\theta, \theta^2]$ with θ equal to the MLE you found in part (c). Prove that the seller's optimal pre-production quantity is $Q^* = 28/3$, if allowing Q to be non-integer. Explain your calculations.

(e) (3 points) If Q must be an integer (the rush productions can still be fractional), what is the optimal decision of Q? Briefly explain your answer.

Solution

(a) **Optimal Rush Production:**
After seeing the realized demand D:

- If $D \leq Q$, the seller has enough pre-produced units. No need for rush production.
- If $D > Q$, the seller should rush produce $D - Q$ units to meet the extra demand.

Since the profit per rush-produced unit is positive ($p > d$), it is always better to produce to meet demand. Thus, the optimal number of rush-produced units is:

$$(D - Q)^+ = \max(D - Q, 0).$$

(b) Either all demands are fulfilled by the pre-productions, or the remaining are fulfilled by rush production.

(c) **MLE for θ:**
Given $D \sim \text{Uniform}[\theta, \theta^2]$, the likelihood is positive only if:

$$\theta \leq \min\{12, 13, 14, 15, 16\} = 12, \quad \text{and} \quad \theta^2 \geq \max\{12, 13, 14, 15, 16\} = 16.$$

Thus, $\theta \in [4, 12]$.
The likelihood function is proportional to $(\theta^2 - \theta)^{-5}$, so to maximize likelihood, we must minimize $\theta^2 - \theta$.
Since $\theta^2 - \theta$ is increasing over $[4, 12]$, the minimum is achieved at $\theta = 4$.
Thus, the MLE is:

$$\boxed{\theta = 4}.$$

(d) **Optimal Pre-Production Quantity Q:**
Now $D \sim \text{Uniform}[4, 16]$. Let Q be the number of units pre-produced. The seller's expected profit is:

$$\pi(Q) = p\,\mathbb{E}[\min(D, Q) + (D - Q)^+] - 5Q - 9\,\mathbb{E}[(D - Q)^+].$$

As $\mathbb{E}[\min(D, Q)] + \mathbb{E}[(D - Q)^+] = \mathbb{E}[D] = 10$, we have

$$\pi(Q) = 10p - 5Q - 9\,\mathbb{E}[(D - Q)^+].$$

Clearly, the above is concave in Q. We also know that the optimal quantity Q^* shall satisfy $Q^* \in [4, 16]$. Next, calculate $\mathbb{E}[(D - Q)^+]$:

$$\mathbb{E}[(D - Q)^+] = \int_Q^{16} (x - Q) \cdot \frac{1}{12}\, dx = \frac{1}{24}(16 - Q)^2.$$

where $\frac{1}{12}$ is the density. Therefore, the expected profit function is:

$$\pi(Q) = 10p - 5Q - \frac{3}{8}(16 - Q)^2.$$

Thus, $Q^* = \frac{28}{3}$.

(e) If Q must be integear, comparing $\pi(9)$ and $\pi(10)$ leads to $Q^* = 9$.